

Bifurcations and the Saddle-Node Mechanism

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Definition: Equilibrium branch

For

$$\dot{x} = f(x; r),$$

an **equilibrium branch** is a curve of points (r, x^*) satisfying

$$f(x^*; r) = 0.$$

Definition: Bifurcation value

A **bifurcation value** is a parameter value at which the qualitative phase portrait changes.

Definition: Bifurcation diagram

A **bifurcation diagram** plots r horizontally and equilibrium values x^* vertically. Time is not an axis of a bifurcation diagram.

Visualization: Branch-style convention

branch style	stability
solid	asymptotically stable
dashed	unstable

Definition: Saddle-node bifurcation

A **saddle-node bifurcation**, also called a **fold**, is the local creation or destruction of two equilibria, one asymptotically stable and one unstable.

Definition: Saddle-node bottleneck

A **saddle-node bottleneck** is slow passage through a neighborhood of the state at which a saddle-node pair collided, after the parameter has moved to the side on which no nearby equilibrium exists.

Definition: Normal form

A **normal form** is a simple parameterized equation representing the leading local dynamics near a bifurcation.

Question: How are equilibria destroyed in the saddle-node normal form?

For

$$\dot{x} = r + x^2,$$

find all equilibrium branches and classify them for every r .

Solution. Equilibria satisfy

$$r + x^2 = 0.$$

Thus

$$\begin{array}{l|l} r < 0 & x_{\pm}^* = \pm\sqrt{-r} \\ r = 0 & x^* = 0 \text{ is a double equilibrium} \\ r > 0 & \text{no equilibrium} \end{array}$$

Since $f_x = 2x$,

$$f_x(-\sqrt{-r}; r) < 0, \quad f_x(\sqrt{-r}; r) > 0.$$

The lower branch is asymptotically stable and the upper branch is unstable. At $r = 0$, $\dot{x} = x^2 \geq 0$, so the origin attracts from the left and repels to the right; it is half-stable. The equilibrium set in the (r, x^*) -plane is the parabola

$$r = -(x^*)^2.$$

Visualization: Saddle-node phase lines and branches

For $\dot{x} = r + x^2$,

r	equilibria	phase line	branch type
$r < 0$	$-\sqrt{-r}, \sqrt{-r}$	$\rightarrow \bullet \leftarrow \circ \rightarrow$	asymptotically stable lower, unstable upper
$r = 0$	0	$\rightarrow \odot \rightarrow$	half-stable fold
$r > 0$	$\emptyset$	$\rightarrow$	no branch

Here $\bullet$ denotes asymptotically stable, $\circ$ unstable, and $\odot$ half-stable.

Question: What changes when the quadratic sign is reversed?

Analyze

$$\dot{x} = r - x^2.$$

Solution. Equilibria exist for $r \geq 0$:

$$x_{\pm}^* = \pm\sqrt{r}.$$

Since $f_x = -2x$,

$$f_x(-\sqrt{r}; r) > 0, \quad f_x(\sqrt{r}; r) < 0.$$

Thus the negative branch is unstable and the positive branch is asymptotically stable. At $r = 0$, $\dot{x} = -x^2 \leq 0$, so the origin repels from the left and attracts from the right. The equilibrium set in the (r, x^*) -plane is the parabola

$$r = (x^*)^2.$$

Question: How is an implicit saddle-node located?

Analyze

$$\dot{x} = r - x - e^{-x}.$$

Solution. Let

$$f(x; r) = r - x - e^{-x}.$$

At a collision of equilibria,

$$f(x; r) = 0, \quad f_x(x; r) = 0.$$

These equations are

$$r - x = e^{-x}, \quad -1 + e^{-x} = 0.$$

Hence

$$x_c = 0, \quad r_c = 1.$$

Moreover,

$$f_{xx}(x; r) = -e^{-x} < 0,$$

so f has a unique global maximum at $x = 0$, with value $r - 1$. Therefore

$$\begin{array}{l|l} r < 1 & \text{no equilibrium} \\ r = 1 & \text{one double, half-stable equilibrium} \\ r > 1 & \text{two equilibria: left unstable, right asymptotically stable} \end{array}$$

Writing $\rho = r - 1$ and using

$$e^{-x} = 1 - x + \frac{x^2}{2} + O(x^3) \quad (x \rightarrow 0)$$

gives

$$\dot{x} = \rho - \frac{x^2}{2} + O(x^3) \quad (x \rightarrow 0),$$

the saddle-node form to leading order.

Proposition: Quadratic reduction at a generic fold

Consider the parameterized equation $\dot{x} = f(x; r)$. Let $f \in C^3$ near (x_c, r_c) and suppose

$$f(x_c; r_c) = 0, \quad f_x(x_c; r_c) = 0, \quad f_r(x_c; r_c) \neq 0, \quad f_{xx}(x_c; r_c) \neq 0.$$

With

$$u = x - x_c, \quad \rho = r - r_c,$$

the local equation is

$$\dot{u} = a\rho + bu^2 + O(|u|^3 + |u\rho| + \rho^2) \quad ((u, \rho) \rightarrow (0, 0)),$$

where

$$a = f_r(x_c; r_c), \quad b = \frac{1}{2}f_{xx}(x_c; r_c).$$

Proof. Taylor expansion gives

$$\begin{aligned} f(x_c + u; r_c + \rho) &= f(x_c; r_c) + f_x(x_c; r_c)u + f_r(x_c; r_c)\rho \\ &\quad + \frac{1}{2}f_{xx}(x_c; r_c)u^2 + O(|u|^3 + |u\rho| + \rho^2) \quad ((u, \rho) \rightarrow (0, 0)). \end{aligned}$$

The first two terms vanish, while $a \neq 0$ and $b \neq 0$. Substituting the definitions of a and b gives the stated equation. $\square$

Corollary: Normal form at a generic fold

Let $a, b \in \mathbb{R} \setminus \{0\}$. The parameterized equation

$$\dot{u} = a\rho + bu^2$$

can be transformed by nonzero rescalings and reversals of state and parameter into

$$\dot{X} = R + X^2 \quad \text{or} \quad \dot{X} = R - X^2.$$

Its equilibrium count is

$$\begin{array}{c|c} -a\rho/b > 0 & 2 \\ -a\rho/b = 0 & 1 \text{ double} \\ -a\rho/b < 0 & 0. \end{array}$$

Proof. The rescaling

$$X = |b|u, \quad R = |b|a\rho$$

gives

$$\dot{X} = \begin{cases} R + X^2, & b > 0, \\ R - X^2, & b < 0. \end{cases}$$

and equilibria satisfy

$$a\rho + bu^2 = 0,$$

which has the stated root counts. $\square$

Question: How long does a saddle-node bottleneck last?

For

$$\dot{x} = \varepsilon + x^2, \quad \varepsilon > 0,$$

find the time required to cross from $-A$ to A , where $A > 0$ is fixed, and determine its asymptotic behavior as $\varepsilon \downarrow 0$.

Solution. Separation gives

$$T_A(\varepsilon) = \int_{-A}^A \frac{dx}{\varepsilon + x^2} = \frac{2}{\sqrt{\varepsilon}} \arctan\left(\frac{A}{\sqrt{\varepsilon}}\right).$$

Since $\arctan(A/\sqrt{\varepsilon}) \rightarrow \pi/2$,

$$T_A(\varepsilon) \sim \frac{\pi}{\sqrt{\varepsilon}} \quad (\varepsilon \downarrow 0).$$

The system has no equilibrium, but it moves slowly near the former collision point $x = 0$.

Question: How is the saddle-node orientation reversed?

For

$$\dot{x} = r + x^2,$$

set

$$y = -x, \quad \rho = -r.$$

Find the equation for y .

Solution. Since $x = -y$,

$$\dot{y} = -\dot{x} = -r - x^2 = \rho - y^2.$$

Thus reversing both the state and parameter axes converts $\dot{x} = r + x^2$ into $\dot{y} = \rho - y^2$.